

SARAH BLAND

Engineer · Designer · Builder

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SKILLS

SOLDERING

FUSION 360

3D CAD

TEAMWORK

PROBLEM SOLVING

ARDUINO

CIRCUIT DESIGN

C++

PYTHON

JAVASCRIPT

July 8, 2026

About Me



Sarah Bland

BOSTON COLLEGE RISING SOPHOMORE

I really enjoy working with design tools and electrical components to build solutions that serve communities in need. Some of my past work includes hosting, marketing, and recruiting for an annual substance abuse charity; building a tool to teach students with disabilities how to count; and designing a tool for an equine therapy center. I am a rising sophomore in Boston College's Human-Centered Engineering Class of 2029. I would like to work in product design after graduating from Boston College, with a focus on hardware or industrial design.

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SKILLS

TREE DISTANCE ALGORITHM FOR PESTICIDE APPLICATION | USDA, BOSTON COLLEGE ENGINEERING

PYTHON

MACHINE-LEARNING

TEAMWORK

COMMUNICATION

PUBLIC-SPEAKING

PROBLEM-SOLVING

Overview

I built and trained a machine learning model in Python to estimate the distance of a tree from a camera, aiming to optimize pesticide application with my partner Michael Busa. We constructed and delivered a presentation on our code within a two-week timeline. As the research this project contributed to has not yet been published, detailed methodology and data are available upon request.

Problem

At present, pesticide application in the United States is either applied by hand or using a set value which doesn't differentiate between tree size, distance or species. This wasted product not only costs significantly more than an optimized approach but can have unintended consequences on nearby populations.

Where is The Lorax when you need him?

How to Teach Distance to a Camera

A regression model to identify the distance a tree is from a camera for improved pesticide application

Sarah Bland and Michael Busa

Contribution

I developed a machine learning model trained on masked tree images cleaned by an algorithm written by my partner Michael Busa. My contribution to this project entailed selecting the best-fit regression model, choosing and assigning weights to heuristics for the model to consider, and assisting Michael in the iteration process for selecting the best tree-masking strategy.

Outcome

My model achieved a Root Mean Squared Error (a measurement of the magnitude of average error of our model) of 0.02 meters, which was two orders of magnitude lower than our intended distance range of 1.5 - 3 meters. Additionally, it reached a Mean Absolute Error of 0.0069 meters.

The Past Solution Graveyard

Failed Solutions

There were more failures, however, these were the ones we thought were most notable.



Grass Line
Inconsistent across dead grass patches



Range Limiter
Attempt to identify tree with grass line



Basic Masks
Too specific; not universally applicable



MATLAB drive
Ghost images, show on browser not on MATLAB

Why They Failed

ITERATION

This is a slide from our presentation which includes a compilation of image-processing techniques which we ultimately abandoned in favor of our present choices.

LESSONS

I learned that iteration is the beginning of success, not the end of it. I also learned the significance of minimal changes to heuristic weights in training ML algorithms.

Digital Projects

CAD-Modeled and 3D-Printed Makeup Tray

FUSION-360

CAD

3D-PRINTING

I created this makeup tray as a simple, clean solution to streamline my daily makeup routine. This tray removed a step from my day of opening and sifting through a makeup bag for what I needed. Before it was implemented, I was using a paper plate for the same purpose, but I decided a tray would allow for ease of use, more space, and aesthetic improvement. I designed the model in Fusion 360 and printed it on a Prusa XL in PLA at a size of approximately 18 cm × 10 cm × 4 cm. Key features include an open top and front for dual access, separated hollow, ribbed sides for content visibility, and a raised base split across four interlocking pieces. I added four interlocking pieces on the bottom because if they were connected to the initial design, the bottom would be completely raised by supports in order to be 3D printed, which would have tarnished the finish and wasted PLA. I also included the open top and front so the tray could be placed at eye level or table height without adjustment of use. I still use it each day, and it saves me about 10 minutes and a gram of frustration a day.



02/18/2026

OpenDots for Braille Literacy | Babson Hackathon 2026

FUSION-360

CAD

BLENDER

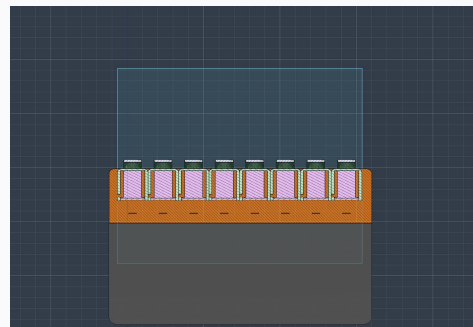
TEAMWORK

PROBLEM-SOLVING

I built the full Fusion 360 model for a digital Braille-style display during Babson University's 2026 five-hour hackathon. There is a rapid decrease in Braille literacy among blind adults despite Braille-literate individuals making up the majority of employed blind individuals, so we created this solution to address the gap in literacy education between visually impaired and able-bodied children. I named each individual solenoid body in the model in Blender so our code could target individual pins in real time. In the physical prototype stage, we would need to incorporate shift registers, which can only map eight Arduino pins at once, so I decided to create the grid in a layout of 8×12 and name them row by column for easy code manipulation. Given the time constraints and the depth of soldering required, we could only create a website, code, and digital model. When a user pressed a letter on our website, a teammate's image-analysis algorithm recognized the character and triggered the corresponding pins in the 3D model, creating a live, interactive Braille output experience. The demo worked and is visible, among other details, in the linked GitHub repository. *My collaborator on this project was Roman Pisani*



04/11/2026



04/11/2026

TideFlow | Personal Scheduling App

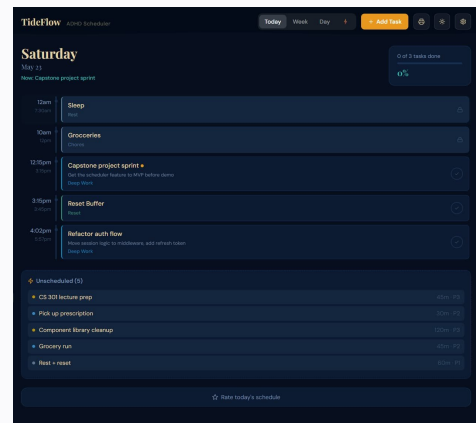
MACHINE-LEARNING

PROMPT-ENGINEERING

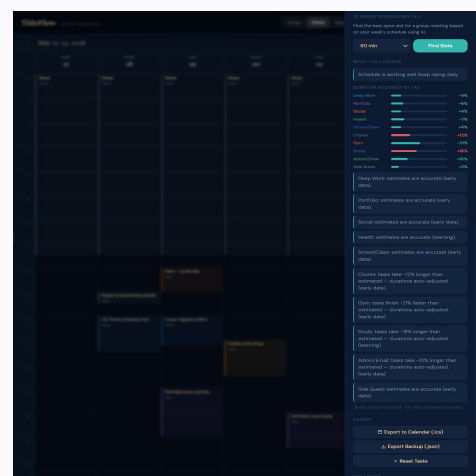
UI-DESIGN

PROBLEM-SOLVING

I built TideFlow, a scheduling app designed around ADHD cognitive patterns, for my brother who was having difficulty managing his schedule. I implemented it using an Object-Oriented Framework in JavaScript. I created a 20-page design spec based on interviews with him and identifying features that would best address his needs. I implemented the system using an iterative rapid-development approach over one week. TideFlow includes a machine learning engine that reads real versus estimated task duration, task satisfaction by category, burnout levels, and when the user prefers to schedule tasks, and it suggests tasks when prompted and provides corrected time estimates for user-added tasks. The model updates per task using recency weighting and satisfaction scores, and it applies safe multipliers to avoid overcorrection. I also built the scheduling logic, which enforces ADHD-friendly rules like protected deep-work windows, nightly cutoffs, burnout-mode load reduction, and collision detection for fixed and flexible tasks. The app includes four views, drag-and-drop rescheduling, ICS export, and a daily survey that feeds back into the ML loop.



05/23/2026



05/23/2026

Poker Chips for the Visually Impaired

FUSION-360

CAD

3D-PRINTING

I designed these poker chips specifically for colorblind players. Since standard poker chip values rely heavily on color differentiation, I introduced four distinct values based on interior geometric cutouts. This approach preserves stackability while allowing for clear tactile and visual differentiation. Each denomination is based on a card suit (such as Clubs, Spades, Hearts, and Diamonds) for immediate recognizability and clean design aesthetic. I modeled the chips in Fusion 360 and 3D-printed them in PLA on a Prusa MINI, with each chip measuring 39mm in diameter and 0.5-1mm in thickness.



03/10/2026

Events

OpenDots | Babson Hackathon 2026

FUSION-360

CAD

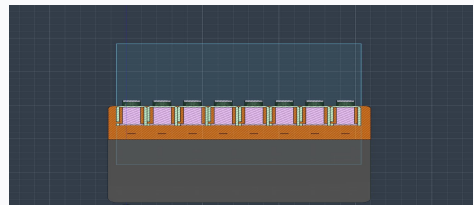
TEAMWORK

PROBLEM-SOLVING

I built the full Fusion 360 model for a digital Braille-style display during Babson University's 2026 five-hour hackathon. Each body in the model was intentionally named so our code could address individual pins in real time. When a user pressed a letter on our website, a teammate's image-analysis algorithm recognized the character and triggered the corresponding pins in the 3D model, creating a live, interactive Braille output experience designed to support early literacy for blind children. More details are available in the linked GitHub repository.



04/11/2026



04/11/2026

RFID Remote Authorization Lock Box | Making The Modern World 2026 Design Conference

CPP

ARDUINO

CIRCUITS

RFID-NFC

SOLDERING

WOODWORKING

3D-PRINTING

FUSION-360

I built this lock box as a smarter alternative to my college's lost and found system for low-to-medium value items. Rather than relying on staff to manually match owners to items, the box integrates directly with the school's existing ID infrastructure. When someone is confirmed as the owner of a lost item, I designed the system to remotely authorize their student ID to unlock the box and retrieve it on their own time. The system runs C++ firmware I wrote for an Arduino Nano, which controls a 5V solenoid lock through a MOSFET switching circuit and communicates with a 13.56MHz RC522 RFID reader over SPI. I handled the card authorization logic and hardware control timing entirely on-device. The custom electronics I wired include flyback protection, gate pull-down resistors, and multi-voltage power distribution (5V/3.3V) for reliable high-current switching. I CAD modeled and 3D-printed a thermal housing for the solenoid to ensure precise actuation alignment and protect nearby components from heat. I also fabricated the enclosure itself-a 19" × 14" × 5" wooden box-using a table saw, adding full cable management and strain relief for a production-quality finish.



11/21/2025



11/21/2025

Portable Counting Game | Exemplary Build Award

CIRCUITPYTHON

LED-MATRIX

SOLDERING

TEAMWORK

PROBLEM-SOLVING

COMMUNICATION

LASER-CUTTING

ADOBE-ILLUSTRATOR

I developed a portable counting game designed for children with combined physical and cognitive impairments, purposefully sizing it to fit directly on a wheelchair tray for independent use. I designed the device to run on a rechargeable battery while retaining plug-in capabilities to ensure uninterrupted sessions. I reduced the interaction to two large buttons, making it accessible to children with limited fine motor control, and added a glass light diffuser above the indicator to soften visual feedback. I soldered and assembled the internal circuitry (a Matrix Portal connected to a HUB75 and mobile battery) and designed the logic controlling the code behind the counting game. I built this in conjunction with Garrett Mackenzie for our Physical Computing class at Boston College.



04/29/2026

TapN | Accelerate@Shea 2026 Cohort

SWIFT

FUSION-360

CAD

3D-PRINTING

ENTREPRENEURSHIP

LEADERSHIP

COMMUNICATION

TEAMWORK

I co-founded TapN, an NFC-based system designed to reduce student phones to only their essential functions during class, minimizing distraction while preserving safety features. I led the hardware design through multiple iterations, building wooden physical mockups, CAD models, and 3D-printed prototypes. I also built object-oriented frameworks in Swift to optimize the tap-to-restrict user flow. When a student enters the classroom, they tap their phone to a physical NFC device which sends a notification to a paired application which restricts all apps not included on a by-school whitelist. This allows for communication tools in times of emergency while simultaneously preventing classroom disruption. Because of our work, TapN was selected for Accelerate@Shea 2026, a competitive startup accelerator at Boston College offering equity-free funding, mentorship, and workshops with industry professionals.



02/17/2026

Keypad-Controlled Vibrating Cat Bed | 2nd Place — Boston College Make-A-Thon 2026

ARDUINO

CIRCUITPYTHON

CIRCUITS

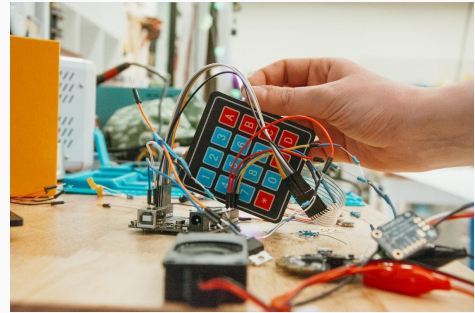
PWM

MOTORS

SOLDERING

PROBLEM-SOLVING

I designed this vibrating cat bed for deaf cats who can no longer experience music audibly, engineering it to deliver calming sensory stimulation through touch. I integrated a wired 4×4 keypad so the owner can easily adjust the bed's settings without disturbing a resting pet. I built the electronics around an Elegoo Uno paired with a CircuitPython Bluefruit, which I programmed to drive a connected speaker and amplifier via PWM signals. I wired five vibration motors through a shared common ground and power line, regulating their intensity with a MOSFET. I mapped the 4×4 keypad to give the user direct control over three functions: power on/off, vibration intensity, and playback of one of three songs, each of which I coded to be felt as rhythmic vibration through the bed's surface.



03/21/2026



03/22/2026

Skills

Skills are followed by the number of listed projects in which they were used.

HARDWARE & ELECTRONICS

SOLDERING (4)

ARDUINO (2)

CIRCUIT DESIGN (2)

RASPBERRY PI (1)

RFID / NFC (1)

LED MATRIX (2)

MOTORS & ACTUATORS (1)

PWM SIGNALS (1)

SOFTWARE & CODE

C++ (1)

PYTHON (3)

JAVASCRIPT (5)

SWIFT (1)

CIRCUITPYTHON (2)

THREE.JS (5)

WEB DEV (5)

PROMPT ENGINEERING (2)

MACHINE LEARNING (4)

DESIGN & MAKING

FUSION 360 (10)

3D CAD (9)

BLENDER (5)

ADOBE-ILLUSTRATOR ILLUSTRATOR (5)

3D PRINTING (4)

UI DESIGN (1)

WOODWORKING (2)

LASER CUTTING (5)

HAND TOOLS (1)

SOFT SKILLS

TEAMWORK (10)

PROBLEM SOLVING (14)

COMMUNICATION (7)

ENTREPRENEURSHIP (1)

LEADERSHIP (2)

EVENT PLANNING (2)

PUBLIC SPEAKING (2)